

USB CLI Reference

BigYellow (Pico car-wash terminal)

Token-based car-wash terminal on RP2040. MIFARE balance is in tokens and drives wash time. Adapter exposes flat top-level `washstart` / `washstop` commands instead of a `wash` dispatcher. Adapter file: application/Source/Adapters/USB_Command_Adapter.c

Source: sevantica_drivers/Source/USB/USB_Command_Handler.c (core) + project adapter application/Source/Adapters/USB_Command_Adapter.c. Adapter commands are dispatched *before* core commands, so an adapter may shadow a core command of the same name.

help / ?

Origin: core

Show available top-level commands (adapter + core).

Command	Description	Example
help	List all registered commands with descriptions.	<code>help</code>
?	Alias of `help`.	<code>?</code>
clear	Clear the terminal screen and re-print the welcome banner.	<code>clear</code>

sys

Origin: core

System-level inspection and lifecycle control.

Command	Description	Example
sys status	Print device ID, site ID, RTC, MIFARE state, free heap, etc.	<code>sys status</code>
sys config	Dump the full live configuration (`Config_PrintToUSB`).	<code>sys config</code>
sys set <param> <value>	Set a single configuration parameter and persist to SD + flash.	<code>sys set mifare.card_init_default_balance 5000</code>
sys version	Show firmware version and build metadata.	<code>sys version</code>
sys reboot	Software reset via watchdog.	<code>sys reboot</code>
sys bootloader	Reboot into RP2040 BOOTSEL (USB mass-storage) mode for re-flashing.	<code>sys bootloader</code>
sys tasks	Print FreeRTOS task list with min-free stack (words).	<code>sys tasks</code>
sys wdt [save erase raw]	Watchdog log: dump latest entries; `save` flushes RAM->flash, `erase` wipes the WDT sector, `raw` dumps the raw flash slots.	<code>sys wdt</code> <code>sys wdt save</code> <code>sys wdt erase</code>

Command	Description	Example
sys security <on off>	Enable / disable the full MIFARE security stack (encryption, HMAC, replay, challenge-response, custom sector keys).	<code>sys security on</code>
sys factory-reset	Reload immutable defaults from the flash factory sector.	<code>sys factory-reset</code>
sys factory-save	Snapshot current source defaults into the flash factory sector.	<code>sys factory-save</code>
sys defaults-apply	Overwrite live (user) settings with the immutable source defaults.	<code>sys defaults-apply</code>

mod

Origin: core

Runtime control of system modules (start / stop / list).

Command	Description	Example
mod list	Print module table with current run state.	<code>mod list</code>
mod start <module>	Start a module. Known names: lcd, mifare, dispenser, buzzer, ioexp, rs485.	<code>mod start dispenser</code>
mod stop <module>	Stop a module by name (same names as `mod start`).	<code>mod stop buzzer</code>

card

Origin: core | Build flag: USE_DRIVERS_NFC

MIFARE card maintenance and recovery (gated by USE_DRIVERS_NFC).

Command	Description	Example
card format	Initialize / re-format the next presented MIFARE card with default data.	<code>card format</code>
card topup <amount>	Add tokens / volume units to the next presented card.	<code>card topup 5000</code>
card log <UID_HEX>	Read and print the on-card transaction log for the given UID.	<code>card log 04A1B2C3D4</code>
card recover <UID_HEX>	Restore card balance from the SD-side recovery log for that UID.	<code>card recover 04A1B2C3D4</code>
card decrypt	Decrypt and dump the next presented card (10 s wait) using current keys.	<code>card decrypt</code>

fs

Origin: core | Build flag: USE_DRIVERS_FATFS

SD card file system access (gated by USE_DRIVERS_FATFS).

Command	Description	Example
fs ls	List files in the SD root and cache them for `fs cat #N` access.	fs ls
fs cat <name #N>	Print contents of a file by name, or by index from the last `fs ls`.	fs cat config.txt fs cat #2
fs find <UID_HEX>	Search SD logs for entries belonging to that card UID.	fs find 04A1B2C3D4
fs format	Delete every file on the SD card (does NOT low-level format).	fs format
fs mount	Expose the SD card to the PC via USB Mass Storage; suspends application I/O.	fs mount
fs unmount	Detach USB MSC and resume application access to the SD card.	fs unmount

wifi (core / ESP8285)

Origin: core | Build flag: USE_DRIVERS_CONNECTIVITY

WiFi via the ESP8285 driver (gated by USE_DRIVERS_CONNECTIVITY).

Command	Description	Example
wifi scan	Trigger an asynchronous scan; results in 3-5 s.	wifi scan
wifi list	Print SSID / RSSI / security from the last scan.	wifi list
wifi status	Show driver state and IP address.	wifi status

rtc

Origin: core

Real-time clock inspection and manual set.

Command	Description	Example
rtc status	Print current time, sync flag, unix timestamp and compile-time fallback.	rtc status
rtc set <YYYY-MM-DD> <HH:MM:SS>	Set RTC date/time and persist to SD.	rtc set 2026-05-04 14:30:00

washstart / washstop (top-level)

Origin: adapter

Manual car-wash control (BigYellow adapter). Note: these are NOT subcommanded, they are top-level commands.

Command	Description	Example
washstart [seconds] [option]	Start a wash without a card. `seconds` <= 3600. `option`: 1=vacuum, 2=brush, 3=pressure (default vacuum).	washstart washstart 120 2
washstop	Stop the active wash immediately.	washstop

Missing / Inconsistent Commands

Cross-checked against: MyWota (Pico water dispenser), Central Control Hub (Pico 2 W master).

`clean` — missing in BigYellow (Pico car-wash terminal)

Present in: MyWota (Pico water dispenser), Central Control Hub (Pico 2 W master).

`dispenser` — missing in BigYellow (Pico car-wash terminal)

Present in: MyWota (Pico water dispenser), Central Control Hub (Pico 2 W master).

`fault` — missing in BigYellow (Pico car-wash terminal)

Present in: MyWota (Pico water dispenser).

`filter` — missing in BigYellow (Pico car-wash terminal)

Present in: Central Control Hub (Pico 2 W master).

`remote` — missing in BigYellow (Pico car-wash terminal)

Present in: Central Control Hub (Pico 2 W master).

`rs485` — missing in BigYellow (Pico car-wash terminal)

Present in: Central Control Hub (Pico 2 W master).

`sdinit` — missing in BigYellow (Pico car-wash terminal)

Present in: Central Control Hub (Pico 2 W master).

`stats` — missing in BigYellow (Pico car-wash terminal)

Present in: Central Control Hub (Pico 2 W master).

`webserver` — missing in BigYellow (Pico car-wash terminal)

Present in: Central Control Hub (Pico 2 W master).

`fault` — missing

MyWota exposes a fault state machine via ``fault status|clear``. BigYellow has no equivalent CLI even though the wash controller can fault.

`clean` — missing

No self-clean concept (water dispenser feature). Acceptable for a car-wash, but worth confirming.

Inconsistent shape: `washstart` / `washstop` vs `wash`

Every other project groups subcommands under a dispatcher. Consider exposing ``wash start [seconds] [option]`` / ``wash stop`` for symmetry with ``dispenser`` and ``clean`` on the other projects.

`stats` / `filter` / `webserver` — missing

Same gap as MyWota — see CCH for the canonical surface.